Assignment 1

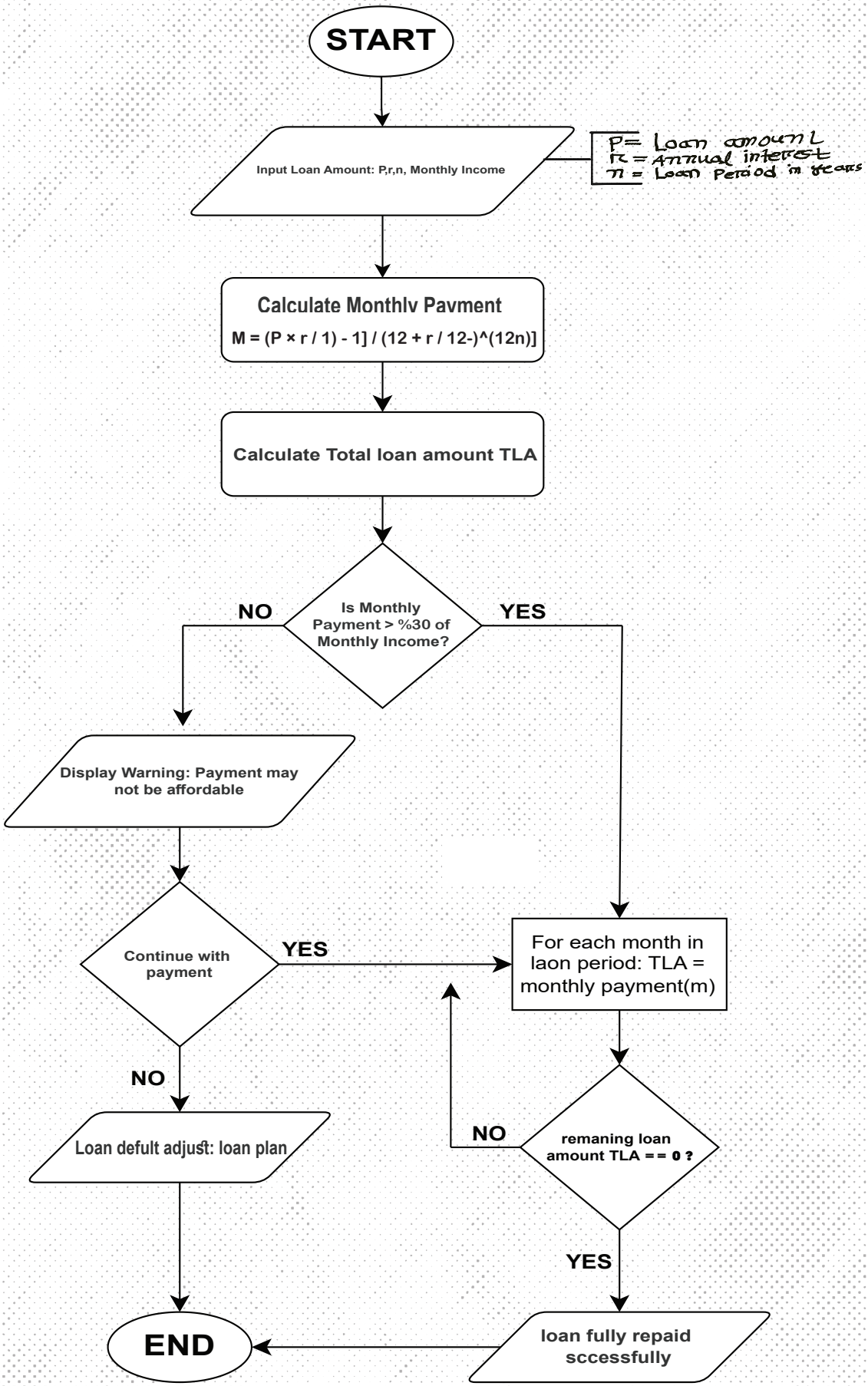
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section 02

FUNDAMENTAL PROGRAMMING CONCEPTS
SCSE1013 :

1. Flowchart Design



2. Explanation of the Flowchart

- **Start:** The process begins.

- **Input:** The system prompts the user to input the loan amount, annual interest rate, loan period in years, and the user's monthly income.
- **Calculate Monthly Payment:** Using the loan amortization formula, the monthly repayment amount is calculated:
  - Formula: $M = (P \times r / 1] / (12 - (1 + r / 12)^{(-12n)}]$
  -
    - M = Monthly payment
    - P = Loan amount
    - r = Annual interest rate (as a decimal)
    - n = Loan period in years

- **Decision (Is Payment > %30 of Monthly Income?):** The system checks if the monthly payment exceeds %30 of the user's income:
  - **Yes:** A warning message is displayed indicating that the payment may not be affordable.
  - **No:** The system proceeds to the next step.

- **Decision (Proceed with Loan?):** The user is asked whether they want to proceed with the loan despite the warning or terminate the process:
  - **Yes:** The system loops through each month, checking the loan status.
  - **No:** The process terminates.

- **Loop Through Each Month:** The loan repayment process is iterated month by month.
- **Decision (Is Loan Fully Paid?):** The system checks if the loan is fully paid:
  - **Yes:** A success message is displayed indicating that the loan repayment is completed.
  - **No:** The system keeps looping until the loan is paid off or the user defaults.
  - **Cannot Repay:** If the loan cannot be repaid, a message is displayed suggesting extending the loan period or adjusting the payment plan.

- **End:** The process ends.

3. Sample Input and Output Cases

Case 1: Affordable Payments

- **Input:**
  - Loan amount: 10,000 USD
  - Annual interest rate: %5
  - Loan period: 3 years
  - User's monthly income: 2,000 USD
- **Calculation:**
  - Monthly payment:

$$M = (10{,}000 \times 0{,}05 / 12) / [1 - (1 + 0{,}05 / 12)^{(-12(36))}] = 299.71 \text{ USD/month}$$

  - %30 of monthly income: 0.3 X 2000 = 600 USD

- **Output:**
  Since the monthly payment of 299.71 USD is less than %30 of the user's monthly income, the loan is affordable. The system proceeds with the loan repayment process.
  Message: "Loan Repayment Successful!"

Case 2: Unaffordable Payments

- **Input:**
  - Loan amount: 20,000 USD
  - Annual interest rate: %10
  - Loan period: 2 years
  - User's monthly income: 1,500 USD
- **Calculation:**
  - Monthly payment:

$$M = (20{,}000 \times 0{,}10 / 12) / [1 - (1 + 0{,}10 / 12)^{(-12(24))}] = 922.49 \text{ USD/month}$$

  - %30 of monthly income: 0.3 X 1500 = 450 USD

- **Output:**
  Since the monthly payment of 922.49 USD is more than %30 of the user's monthly income, a warning message is displayed. The user is asked whether they want to proceed with the loan or terminate the process. If the user decides to proceed, the loan may result in default unless adjusted.
  Message: "Warning: Payment may not be affordable. Loan Default: Extend Loan Period or Adjust Payment Plan."